

Patience Nuhvete Joshua

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RESEARCH PROFILE

M.Tech (AI/ML) student working on visual representation learning and Vision Transformers. I study robustness and generalization under distribution shift, using synthetic-image detection as a testbed for how model capacity affects transfer. I have two manuscripts under review and am interested in foundation-model representations, efficient adaptation, and rigorous evaluation.

Research interests: visual representation learning; Vision Transformers; robustness and generalization under distribution shift; model capacity and efficiency; adaptation of foundation models.

EDUCATION

M.Tech, Computer Science & Engineering (AI/ML)

Expected 2027

Shobhit Institute of Engineering & Technology, Meerut, India

CGPA: 8.96/10 (after two semesters)

Supervisor: Dr. Ruchi Holker

B.Sc., Computer Science

2025

Taraba State University, Jalingo, Nigeria

Second Class Upper Division, CGPA: 3.81/5

RESEARCH EXPERIENCE

Research Assistant / AI Researcher

2026 – Present

Shobhit Institute of Engineering & Technology, Meerut, India | Supervisor: Dr. Ruchi Holker

- Investigated the relationship between Vision Transformer capacity and cross-distribution robustness by fine-tuning DeiT-Tiny (~5.5M parameters) and ViT-Base (~85.8M parameters) on GAN- and diffusion-generated face images and testing transfer in both directions.
- Implemented the training and evaluation pipeline in PyTorch and timm (AdamW), running multiple random seeds and reporting ROC-AUC separately for in-distribution and cross-distribution test sets to measure the generalization gap.
- Found near-saturated in-distribution ROC-AUC (>0.998) for both models but cross-distribution ROC-AUC of only 0.51–0.54; reported the effect of capacity on transfer as inconclusive and documented the study's limitations.
- Reviewed CNN, hybrid CNN–Transformer, and ViT-based detectors (38 references) and proposed a conceptual framework relating generator diversity, model capacity, and representation invariance, with leave-one-generator-out evaluation as the protocol for testing it.

MANUSCRIPTS

[1] P. N. Joshua, R. Holker, E. I. Simon. "A Comparative Review of Vision Transformer Architectures for Synthetic Image Detection: From Local Artifacts to Global Inconsistencies." Manuscript submitted to *Signal, Image and Video Processing* (Springer); under review.

[2] P. N. Joshua, R. Holker. "Capacity vs. Generalization: A Preliminary Study of Vision Transformer Capacity and Cross-Distribution Robustness in Deepfake Detection." Manuscript submitted to ICISCS 2027; under review.

SELECTED PROJECTS

Frequency–Spatial Fusion for Generalizable Deepfake Detection (research proposal)

2026

- Surveyed 30 papers on frequency-domain methods (FFT, DCT, wavelet), hybrid CNN–Transformer detectors, and cross-dataset generalization; wrote a gap analysis that framed the proposal.

CNN-to-ViT Architectural Study and Seminar

- Studied the shift from convolutional to Vision Transformer and hybrid CNN–ViT designs, and presented the findings in a 30-slide departmental seminar.

Predictive Modeling Projects, GCI World 2026

2026

- Compared Logistic Regression, Random Forest, and CatBoost for telecom churn prediction and selected CatBoost on ROC-AUC.
- Built an XGBoost/LightGBM ensemble for an NFL Draft prediction competition and reverted feature additions that lowered the leaderboard score.

TECHNICAL SKILLS

Programming: Python; HTML, CSS, JavaScript, PHP, SQL (MySQL)

ML / Deep Learning: PyTorch, timm, TensorFlow, scikit-learn, XGBoost, LightGBM, CatBoost, pandas, NumPy

Research Areas: Computer vision, Vision Transformers, visual representation learning, transfer learning and fine-tuning, robustness under distribution shift, model evaluation (ROC-AUC, multi-seed protocols)

Tools: Git/GitHub, Jupyter, Google Colab, Streamlit

Languages: English, Hausa

ADDITIONAL EXPERIENCE

Intern, Sandlip Digital Consulting

Jan 2024 – May 2024

- Supported software development using HTML, CSS, JavaScript, PHP, and MySQL.

Technology Volunteer, Tech Girls Drive (Tech4Dev), Taraba, Nigeria

Oct 2025

- Taught computer skills to pupils alongside the Marben Foundation.

Instructor, vMix livestreaming training class

- Taught livestreaming to more than 40 participants.